# Juahn Jeong

Senior Backend Engineer / Tech Lead

johnnyapu15@gmail.com


Backend engineer leading an engineering team building AI and 3D services at Bucketplace, driving new service launches, high-traffic reliability, in-app subscription and affiliate monetization, and internal shared platform development. Recognized with the company's **Eng Award**, **AI Award**, and **AI Native MVP** for leading key technical initiatives and improving organizational productivity. Improved API and loading performance for the Room Planner service and automated the 3D asset production, review, and deployment pipeline, contributing to approximately 7x WAU growth and 6x annual GMV growth (KRW 860 million) while reducing asset production costs by 88%. Subsequently led backend development for the global AI service (Ohouse AI), launching it in 4 weeks and building in-app subscription billing and Amazon affiliate product recommendations. Building on experience developing and operating the Room Planner AI Interior Assistant, led the development of the internal shared **AI Agent Platform**. Narrowed the initial platform scope to essential shared capabilities to build team consensus, and directly handled everything from core runtime server development to onboarding another team's natural language search service (AI Curator) and verifying pre-launch performance and scalability.

## Experience

### Bucketplace

Aug 2022 - Present

Senior Backend Engineer / Tech Lead

Lead an engineering team building AI and 3D services, responsible for product and platform design, development prioritization, hands-on core server implementation, and collaboration with FE, ML, and partner teams.

- **Led AI Agent Platform Development**: Defined the platform development scope and built team consensus to standardize agent execution and conversation management features previously built separately by each team. Directly developed the shared runtime server and migrated existing chat agent services to the platform architecture.
- **AI Curator Platform Integration & Launch Readiness**: Applied the shared platform to another team's natural language search service (AI Curator) and integrated real-time conversation handling and history management. Established server scaling criteria and operational configurations through pre-launch load testing, and resolved latency issues when retrieving long conversations.
- **AI Interior Assistant Development & Evaluation Environment**: Developed the Room Planner AI chat server providing product search, image synthesis, and style recommendations, and led the spatial agent MVP from defining client integration specifications to server implementation and deployment. Built automated evaluation and conversation replay tools to check conversation quality before deployment.
- **Ohouse AI Global Launch & Subscription/Affiliate Monetization**: Built the backend and asynchronous image generation structure for the global AI interior service, completing a targeted 6-week launch schedule in 4 weeks. Developed the Apple and Google in-app subscription billing server and integrated Amazon affiliate product recommendations, recording a 10.5% purchase conversion rate after affiliate product clicks.
- **Room Planner Performance, Asset Automation & AR Improvements**: Reduced API response time (200ms to 80ms) and loading time (10s to 2.5s), and maintained 99.95% availability under 10.2x normal traffic, contributing to approximately 7x WAU growth and 6x annual GMV growth (KRW 860 million). Automated the 3D asset production, review, and deployment process to increase monthly output from 68 to 800 assets, reducing production costs by 88% and saving over 40 operational days annually, receiving the **Eng Award**. Also contributed to a 1.7x increase in purchase conversion rate by reducing AR model size by 90% and introducing Size Box to expand supported products from 2,210 to 231,000.
- **Team Enablement & Productivity Improvements**: Led backend onboarding and code reviews for new hires and cross-functional engineers, and maintained development momentum during organizational restructuring by running design reviews and technical meetings. Built AI-based release note generation and data analysis tools to reduce repetitive work, receiving the **AI Award** and **AI Native MVP**.

### LG CNS

Jan 2020 - Jul 2022

System & Solution Programming Consultant

- Designed, developed, and operated a real-time multi-user editing state synchronization server and backend APIs for Unity client integration.
- Automated manual build, test, and deployment processes into a CI/CD pipeline to reduce deployment time.

## Technical Skills

**Languages & Frameworks**: Kotlin, Python, TypeScript / Spring Boot, Node.js
**Data & Infrastructure**: MySQL, MongoDB, Redis, Valkey, Kafka / AWS, Kubernetes, KEDA

## Education

**University of Seoul** | M.S. in Computer Science | Research on AI system performance optimization — 2018 - 2019

**University of Seoul** | B.S. in Computer Science — 2012 - 2017

## Key Experience

### 1. Internal Shared AI Agent Platform Development & Cross-Team Adoption

**Background**

Multiple internal teams preparing AI agent services were duplicating conversation history management and real-time response handling features. Meanwhile, the initial platform plan had a broad scope that made it difficult to deliver quickly in time for actual service schedules.

**Role**

Led platform development and narrowed the phase-one scope to execution, session, and streaming features immediately needed for service integration, building team consensus. Directly developed the shared runtime server to migrate the existing Interior Assistant agent to the platform first, and then led the integration for another team's AI Curator service to use the platform's conversation storage and response handling flow.

**Result**

Migrated the existing production agent service to the internal platform and completed integration for another team's new service on the same platform, enabling agent features to be developed and operated on a shared architecture.

### 2. AI Curator Pre-Launch Performance Verification & Operational Readiness

**Background**

Ahead of launching AI Curator, another team's natural language search service, there were no server scaling criteria suited to agent services with long external model wait times, and longer conversations caused response latency and ordering mismatches.

**Role**

Took charge of pre-launch backend stabilization, setting up a load testing environment matching realistic service conditions to identify per-server processing limits. Configured auto-scaling criteria and operational settings based on pending workload, and identified and fixed causes of response ordering issues and latency when retrieving long conversations found during integration.

**Result**

Resolved the retrieval latency and response ordering issues, and completed launch readiness by applying the scaling criteria and operational configurations verified through platform integration and performance testing.

### 3. Room Planner Performance Improvement & Asset Pipeline Automation

**Background**

Slow initial entry times in the mobile 3D interior service were causing user drop-offs, while the system also needed to handle traffic spikes during major promotions. Additionally, expanding placeable products required a large volume of 3D assets, but outsourced production, review, and deployment were handled manually, requiring higher output and lower costs.

**Role**

Directly resolved bottlenecks in initial entry APIs and 3D model loading, and prepared the environment for high-volume traffic. Built an automated pipeline covering the entire manual 3D asset production, review, and deployment process, and developed AI-based quality classification and a web operations dashboard so the operations team could review and deploy assets without developer support.

**Result**

Reduced API response time (200ms to 80ms, payload size reduced by 54%) and loading time (10s to 2.5s), and maintained 99.95% availability under 10.2x traffic, contributing to approximately 7x WAU growth and 6x annual GMV growth (KRW 860 million). Increased monthly asset output from 68 to 800, cut production costs by 88%, and saved over 40 operational days per year, earning the **Eng Award**.